METRICS

fβ - score

$$f\beta = \frac{1+\beta^2}{\frac{1}{\text{Precision}} + \frac{\beta^2}{\text{Recall}}}$$

$\beta < 1$: favours precision

$\beta > 1$: favours recall

$$\text{f1 - score} = \frac{TP}{TP + \frac{1}{2}(FP+FN)}$$

$$\text{Precision} = \frac{TP}{TP+FP} \qquad \text{Recall} = \frac{TP}{TP+FN} \qquad \text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$$

Macro-f1 score: only for classes with the same no. of members

Micro-f1

We not everything.

act \ pred		
	TP	FN
	FP	TN

$$\text{Micro-f1} = \frac{\text{net TP}}{\text{net TP} + \frac{1}{2}(\text{net FN} + \text{net FP})} = \frac{\sum_i C_{ii}}{\sum_i C_{ii} + \frac{1}{2}\left(\sum_{i \neq j} C_{ij} + \sum_{i \neq j} C_{ji}\right)}$$

For binary class dataset, micro-f1 = accuracy.

ROC - AUC

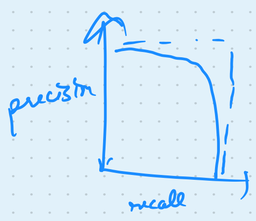
Graph of TPR vs FPR → $\frac{TP}{TP+FN}$, $\frac{FP}{TP+FP}$

sensitivity / recall ; 1 - specificity

PR - AUC

Graph of precision vs recall → $\frac{TP}{TP+FP}$, $\frac{TP}{TP+FN}$

- Works better for imbalanced coz it uses only TPR, not FPR



Weighted F1

$$\text{weighted F1} = \sum_{i \in I}^{N} w_i (F1)_i \qquad \text{where } w_i = \frac{n(\text{samples of } i)}{N}$$

Balanced accuracy

Mean of sensitivity (recall, hit rate, TPR) and specificity (selectivity, TNR)

$$\text{Bal. accuracy} = 0.5\left(\frac{TP}{TP+FN} + \frac{TN}{FP+TN}\right)$$

Used when negative class also matters (f1 gives TN zero importance)

Matthew Correl. Coeff

Essentially take correl b/w true & predicted values.

Higher = better

$$\phi / MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

'matthews_corrcoef'

Cohen's Kappa → multiclass: how much better agreements are over chance

		Professor A		
		Accept	WL	Reject
Professor B	Accept	4	6	3
	WL	1	2	0
	Reject	1	2	6

$$P(\text{Agree}) = \frac{4+2+6}{25} = 0.48$$

Now,

$$P(\text{AcceptA}) = \frac{4+1+1}{25} = 0.24$$

$$P(\text{AcceptB}) = \frac{4+6+3}{25} = 0.52$$

For both to agree by chance:

$$\text{ChanceAgree(Accept)} = P(\text{AcceptA}) \times P(\text{AcceptB}) = 0.1248$$

Likely $\text{ChanceAgree(WL)} = \left(\frac{6+2+2}{25}\right)\left(\frac{1+2+0}{25}\right) = 0.048$

$\text{ChanceAgree(Reject)} = \left(\frac{3+0+6}{25}\right)\left(\frac{1+2+6}{25}\right) = 0.1188$

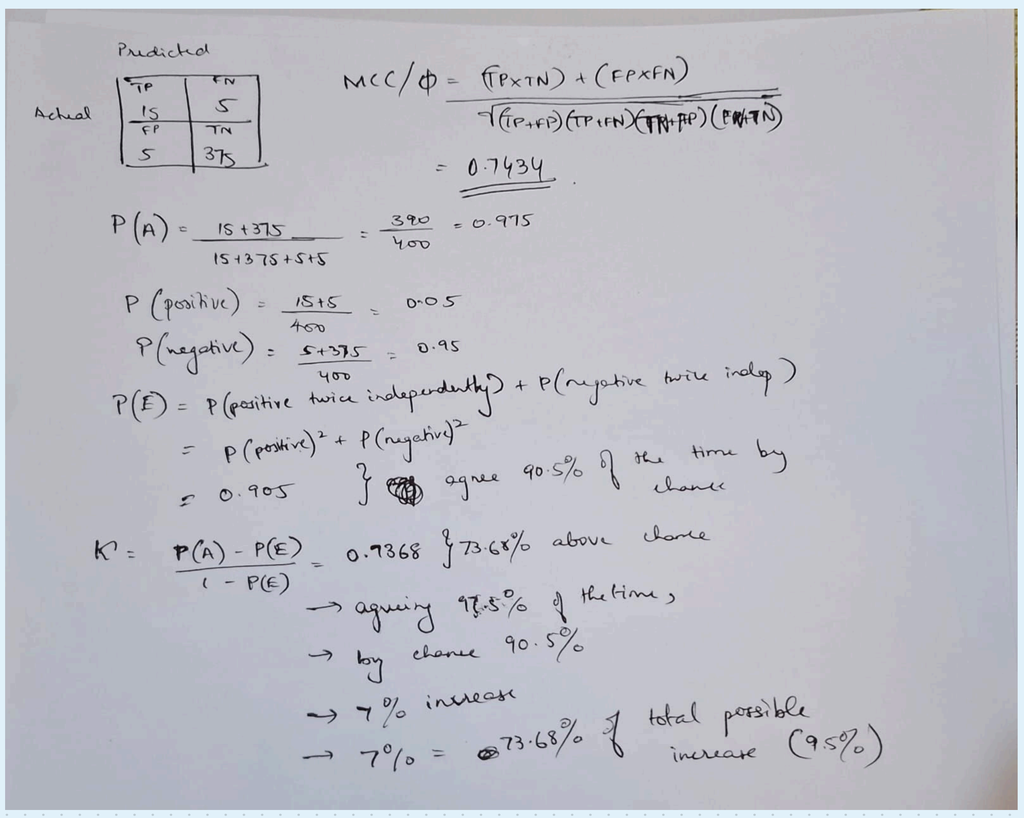
Sum:

$\text{ChanceAgree} = 0.3024$

$$\text{K-score} = \frac{\text{Agree} - \text{ChanceAgree}}{1 - \text{ChanceAgree}} = 0.2545$$

Agree / 1 → remove chanceAgree from both. Kinda like avg.

≤ 0 → poor
0.01 - 0.20 → slight
0.21 - 0.40 → fair
0.41 - 0.60 → moderate
0.61 - 0.80 → substantial
0.81 - 1.00 → v perfect



SAMPLING

- probability
  - simple random
  - systematic (skip k after ordering)
  - stratified
  - cluster
- non-probability
  - convenience
  - quota
  - judgement
  - snowball

OVERSAMPLING

Random

duplicate datapoints from minority

overfitting — simple

Synthetic Minority Oversampling Technique (SMOTE)

$$O_{new} = O + z(N-O)$$

pick instance of minority at random → origin O

find nearest neighbours of minority (KNN)

pick one at random, draw line to it / pick all and draw lines

pick random factor $z \in [0,1]$

place point on line $(z \times 100)\%$ away from O

new samples, not duplicates → density of minority class in one specific region high

↓ overfitting

→ new points created w/o considering majority class - if minority overlaps, causes ambiguity

Borderline SMOTE

Any given point:
- Noise: all neighbours from different class
- Border: mix of majority, minority
- safe: all minority

- Origin sampling done only with Border points.

Borderline SMOTE 1: interpolate b/w O and nearest minority only

" " 2: " " " " " minority and majority

K-Means SMOTE

1. Do k-means clustering
2. select clusters with high proportion of minority
3. Do SMOTE: more sparse clusters, generate more points

inherits all weaknesses of k means
- select k
- uniform size cluster, spherical
- ignores global behaviours
- sensitive to outliers

SVM SMOTE

1. Train SVM on data
2. Identify support vectors for minority class
3. Sample O from these support vectors
4. Find KNN → either minority or majority
5. majority of N ∈ majority: interpolate b/w O and N
   majority of N ∈ minority: extrapolate beyond decision boundary

$$O = O + z(O-N)$$

ADASYN: Adaptive Synthetic Sampling

Adapts to regions where imbalance is severe

1. Select point A from minority
2. Difficulty ratio (DR) = $\frac{n(\text{majority N})}{n(\text{minority N})}$
3. More samples generated for higher DR

Repeat -

1. DR → given
2. weight $w_i = \frac{DR_i}{\sum DR}$ for each sample
3. total samples to be generated
   $G = (N_{maj} - N_{min})\beta$
   $\beta \in [0, 1]$ → how much to balance. typ. 0.5
4. $G_i = w_i G$
5. Generate G_i samples using SMOTE